

FIELD VISIT DATA COLLECTION FORMAT for MULTIDISCIPLINARY PROJECT

[Phase I: Problem Identification (2nd to 3rd Semester Transition)]

1. Individual Student Information

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2. Field Visit Details

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet? (Name, Role of the person, Phone #, Email ID (if applicable))	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
3/10/25	5.Rural Development and Smart Village	Dr.K.V.Prakash	GKVK	Yes

Note: The individuals you interact with may belong to either the organised sector or the unorganised sector, for example, a fisherman, farmer, local shopkeeper, or a retired Anganwadi worker.

3. Person(s) Visited Interaction Summary (Data Collection)

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
Dr KV Prakash	Proffesor	Organised	Personal Meeting

Key Issues Highlighted: Farmers face major crop losses due to slow, costly, and manual disease detection, lacking quick and affordable solutions for early identification and control

4. Observations & Data Collected

Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches/Documents Attached
Around 20–40% of global crop yield is lost annually due to pests and diseases. Over 500 million small-scale farmers face productivity loss from late disease detection. Manual diagnosis can take 2–5 days and cost up to ₹500–₹2000 per sample. Early detection could reduce crop loss by up to 30% and increase income significantly.	Delayed detection of plant diseases. High cost and inaccuracy of manual diagnosis. Limited access to expert help in rural areas. Reduced crop yield and farmer income. Lack of awareness about preventive measures.	Yes  

Note: Quantitative data for social issues refers to measurable, numerical information that helps to objectively understand the scale, impact, and urgency of a problem. (Ex: Number of people affected; Economic loss; Resource wastage; Time lost; Frequency of an issue; Health data etc.)

5. Problem Statement Draft (Tentative)

Draft Problem Statement(s)

Farmers are facing major crop losses due to insect attacks and delayed identification of plant health issues. The existing diagnostic methods are manual, time-consuming, and costly, making it difficult for farmers especially in rural areas to detect problems early and take timely action. There is an urgent need for a quick, affordable, and accurate solution to identify plant diseases and suggest effective remedies to improve agricultural productivity and sustainability

Note: Write a clear and short problem statement about a real issue, using what you observed during the visit and what you learned from the people you spoke to.

6. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	Justification (How the problem aligns)
SDG 2: Zero Hunger – Promotes sustainable agriculture and food security by reducing crop loss.	The project contributes to sustainable farming by helping farmers detect diseases early, increase yield, and reduce wasteful resource use—supporting food security, innovation, and environmental protection
SDG 9: Industry, Innovation and Infrastructure – Encourages innovative AI-based solutions in agriculture.	
SDG 12: Responsible Consumption and Production – Supports efficient use of resources and minimizes waste.	
SDG 15: Life on Land – Protects ecosystems by reducing harmful pesticide use through targeted treatment.	

SUMMARY OF THE SUSTAINABLE DEVELOPMENT GOALS

<u>SDG Sector</u>	<u>Focus Areas</u>	<u>SDG Sector</u>	<u>Focus Areas</u>
(1)Healthcare & Public Health	• Government & Private Hospitals • Primary Health Centre's (PHCs) • Rural & Urban Health Clinics • Rehabilitation Centre's • Biomedical Research Institutes • Blood Banks & Diagnostic Labs • Medical Device Manufacturing Units	(7) Education & EdTech Initiatives	• Rural & Government Schools • Skill Development & Vocational Training Centre's • Educational NGOs for Underprivileged Children • Libraries & E-Learning Hubs • Special Needs & Inclusive Education Centres
(2)Waste Management & Environmental Conservation	• Municipal Waste Collection & Segregation Units • Solid Waste Management Plants • Hazardous Waste Disposal Centre's • Air & Water Pollution Monitoring Centre's • Water Treatment Plants • Renewable Energy Farms • Forest & Wildlife Conservation Centre's	(8)Urban structure & Transportation	• Metro Rail & Public Transport Hubs • Traffic Management & Smart City Projects • Road Safety & Accident Prevention Centre's • Smart Parking & Electric Vehicle (EV) Infrastructure • Smart City Projects • Toll Plazas & NHA
(3)Water Resource Management	• Water Treatment & Sewage Plants • River Conservation & Irrigation Projects • Groundwater Recharge Initiatives • Coastal & Marine Conservation Programs	(9) Housing & Smart Infrastructure	• Affordable Housing & Slum Rehabilitation Projects • Sustainable & Green Building Projects • Smart Housing for the Elderly & Disabled • Urban & Rural Infrastructure Development Authorities
(4)Food & Nutrition Security	• Food processing & preservation units • Public Distribution System (PDS) Centre's • Mid-Day Meal Programs in Schools • Agricultural Cooperatives & Farmers' Markets	(10) Cybersecurity & Digital Governance	• Cybercrime & Forensic Labs • Digital India & E-Governance Initiatives • Aadhaar & Digital Payment Centre's • Smart Policing & AI-Based Surveillance Units
(5)Rural Development & Smart Villages	• Model Smart Villages • Rural Skill Development Centre's • Agri-Tech & Precision Farming Centre's • Rural non-conventional electrification projects • Agricultural Research centre's Dairy Farms & Veterinary Centre's • Food Processing Units	(11) Renewable Energy & Sustainability	• Solar & Wind Energy Farms • Hydroelectric Power Stations • Waste-to-Energy Plants • Recycling & E-Waste Management Units • Smart Grid & Energy Conservation Projects
(6)Disaster Management & Emergency Services	• National Disaster Response Force (NDRF) Training Centre's • Fire & Safety Departments • Emergency Control Rooms (Floods, Earthquakes, Cyclones) • Flood-Prone & Landslide-Affected Areas	(12) Law Enforcement & Governance	• Cybercrime Cells • Local Administrative Offices • Courts & Legal Aid Centre's
(13) Special Economic Zones & Industry Clusters		• MSMEs & Startups, Rural Cooperatives • Large-Scale Manufacturing Units	

STANDARD OPERATING PROCEDURE FOR FIELD VISIT(s)

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Objective: To equip engineering students with the ability to recognize socially impactful issues during their summer break by immersing themselves in their local environments, observing ground-level realities, and engaging with communities. These issues are to be addressed through logical analysis and technological innovation, fostering socially responsible engineering solutions aligned with the Sustainable Development Goals (SDGs).

1. Pre-Visit Preparation (*Communication Channel*)

- All students must ensure they have an official atria.edu email ID (available through the respective Department HoDs).
- All program-related communications, updates, and logins will be managed exclusively via this institutional email only (Personal email ID is not considered).
- Students will be enrolled on the collaborative platform ‘**CommunityX**’ to enable peer interaction, expert support, and resource sharing during the summer engagement.

2. Identifying the Project Area

- Atria Institute of Technology provides a summarized version of the UN’s Sustainable Development Goals (SDGs), which will be shared with students (official Email ID).
- Students are required to closely observe their surroundings and identify **social or environmental issues** affecting daily life in their communities.
- The identified issue must:
 - Be **relevant to society**
 - Have a **potential user base**
 - Offer scope for **technological or process-based intervention**
- Select a problem area that aligns with one or more SDG goals, ensuring that the issue can be approached from an engineering perspective.

3. Prior Research & Planning

- Conduct desk research or connect with individuals who know the chosen domain.
- Prepare a structured set of **interview questions** and **observation checklists**.
- Schedule meetings or visits with local stakeholders (officials, field engineers, community leaders, NGOs, etc.)

4. Travel & Permissions

- Get formal approval from the Institute SIP Coordinators, Dr. Mydhili, Associate Dean, and Prof. Somesh (ECE) through email (if required).
- Plan field visits responsibly, ensure you carry identification, notepads/digital tablets, safety precautions, and ethical protocols.

5. During the Visit: Data Collection

- Use the provided field visit data collection template to record:
 - Date and Location
 - Sector visited and its relevance
 - Stakeholder details and their feedback
 - Observed technical/social/economic challenges
 - Measurable **data wherever applicable**
 - **Visual/audio** documentation (**photos, sketches, voice notes, etc.**)

6. Post-Visit Submissions

Within **7 days of the 3rd semester commencement**, submit the following:

Deliverable	Description
Field Visit Report	Follows official format: includes visit summary, challenges observed, photos
Stakeholder Log	Details of each interaction: who, what discussed, insights gained
Problem Statement Draft	Clear articulation of the identified problem/ issue
SDG Mapping	Indicate relevant Sustainable Development Goal(s) linked to the issue
Attachments	Media files, documents, observation forms, supporting documents

7. Recommended Timeline

Vacation Week	Activity
Vacation Week 1 & 2	Sector selection and scheduling
Vacation Week 3-5	Field visit and data collection
Week 1 of 3rd Sem	Submit reports and problem statements for review

8. Help & Assistance:

- Register on the **Xcelerator platform** before initiating any work.
- Use **CommunityX** for assistance with scheduling, clarifications, mentoring, and peer discussion.
- Team formation will occur post-submission, based on overlapping interest areas. Each team will receive guidance from faculty mentors, alumni, and industry professionals.

9. Code of Conduct and Points to Note:

- Dress professionally, wear an Atria ID Card and conduct yourself respectfully at all times.
- Take consent before capturing any photo, video, or recording.
- Do not make promises or raise expectations during interactions with community members.
- Ethical and respectful use of information and data is mandatory.
- Students may explore more than one issue and are welcome to submit multiple problem statements.
- Follow the schedule strictly to ensure timely evaluation and progression.

Note: For any further information, you may contact:

Dr. Mydhili, Associate Dean - mydhili.nair@atria.edu

Prof. Somesh (ECE) - bs.somesh@atria.edu